Q2. Write a program for process creation using C

- **Orphan Process**

```
  GNU nano 7.2
#include <stdio.h>
#include <unistd.h>

int main() {
    if (fork() > 0) {
        printf("Parent exiting\n");
    } else {
        sleep(5);
        printf("Child running\n");
    }
    return 0;
}
```

```
ishika@DESKTOP-MF8PUSJ:~$ nano orphan.c
ishika@DESKTOP-MF8PUSJ:~$ gcc orphan.c
ishika@DESKTOP-MF8PUSJ:~$ ./a.out
Parent exiting
ishika@DESKTOP-MF8PUSJ:~$ Child running
```

- **Zombie Process**

```
  GNU nano 7.2
#include <stdio.h>
#include <unistd.h>

int main() {
    if (fork() == 0) {
        printf("Child process exiting\n");
    } else {
        sleep(10);
        printf("Parent process running\n");
    }
    return 0;
}
```

```
sudo apt install gcc
ishika@DESKTOP-MF8PUSJ:~$ gcc fork.c./a.out
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
ishika@DESKTOP-MF8PUSJ:~$ sudo apt update
[sudo] password for ishika:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1410 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [230 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [9820 B]
Get:14 http://archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [929 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1720 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [317 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.0 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1525 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [312 kB]
Get:24 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [212 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:26 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.3 kB]
Get:27 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [31.8 kB]
Get:28 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [19.8 kB]
Get:29 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2547 kB]
Get:30 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2369 kB]
Get:31 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [583 kB]
Get:32 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:33 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [556 B]
Get:34 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [32.1 kB]
Get:35 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [6816 B]
Get:36 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:37 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [496 B]
Get:38 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40.4 kB]
```

```
ishika@DESKTOP-MF8PUSJ:~$ sudo apt install build-essential
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
```

```
ishika@DESKTOP-MF8PUSJ:~$ ls
fork.c  zombie.c
ishika@DESKTOP-MF8PUSJ:~$ gcc zombie.c
ishika@DESKTOP-MF8PUSJ:~$ ./a.out
Child process exiting
Parent process running
ishika@DESKTOP-MF8PUSJ:~$
```